Methods and Final Review Lab
[No Submission]

CSE110: Programming Language I

You must not hard code any of the tasks, which means your code should work for any valid user input.

Methods

1. [A,B,C,D should be written in a single java file]

A. Write a method called **evenChecker** that takes an **integer** number as its argument and prints whether the number is even or odd **inside the method**.

Sample Method Call	Sample Output
`evenChecker(10);`	`Even!!`
`evenChecker(17);`	`Odd!!`

B. Write a method called **isEven** that takes an **integer** number as an argument and **returns** boolean true if the number is even otherwise **returns** boolean false.

Sample Method Call	Sample Output
`boolean result = isEven(10);` `System.out.println( result );`	`true`
`boolean result = isEven(17);` `System.out.println( result );`	`false`

C. Write a method called **isPos** that takes an **integer** number as an argument and **returns** boolean true if the number is positive otherwise **returns** boolean false.

Sample Method Call	Sample Output
`boolean result = isPos(-5);` `System.out.println( result );`	`false`
`boolean result = isPos(12);` `System.out.println( result );`	`true`

D. Write a method called **sequence()** that takes an **integer** in its parameter called n. Now, if n is **positive** then it prints all the **even** numbers from **0 to n**, otherwise if n is **negative** it prints all the **odd** numbers from **n to -1**.

Note: You must call the methods from **CW-1B** and **CW-1C**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output	Explanation
`sequence(10);`	`0 2 4 6 8 10`	Here, 10 is positive so 0,2,4,6,8,10 were printed.
`sequence(-7);`	`-7 -5 -3 -1`	Here, -7 is negative so -7,-5,-3,-1 were printed.
`sequence(7);`	`0 2 4 6`	Here, 7 is positive so 0,2,4,6 were printed
`sequence(-8);`	`-7 -5 -3 -1`	Here, -8 is negative so -7,-5,-3,-1 were printed.

2. [A,B,C should be written in a single java file]

A. Write a method called **circleArea** that takes an **integer** radius in its parameter and **returns** the **area** of the circle.

Note: area of a circle is πr^2

Sample Method Call	Sample Output
`double area = circleArea(5);` `System.out.println(area);`	`78.5398`

B. Write a method called **sphereVolume** that takes an **integer** radius in its parameter and **returns** the **volume** of the sphere.

Note: volume of a sphere is $\frac{4}{3}\pi r^3$

Sample Method Call	Sample Output
`double volume = sphereVolume(5);` `System.out.println(volume);`	`523.5987`

C. Write a method called **findSpace** that takes two values in its parameters one is an **integer** diameter and another one is a String. Using the given diameter, this method should calculate the Area of a circle or the Volume of a sphere depending on the value of the second parameter. Finally, it should print the result **inside the method**.

Note: **You must call** the method written in task **CW-2A & CW-2B**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output
`findSpace(10,"circle");`	`78.5398`
`findSpace(5,"sphere");`	`33.5103`
`findSpace(10,"square");`	`“Wrong Parameter”`

3. Write a method called **isPrime** which takes an integer in its parameter to check whether a number is prime or not. If the number is prime then the method returns boolean **true** otherwise it returns boolean **false**.

Sample Input	Sample Output
`boolean check = isPrime(7);` `System.out.println(check);`	`true`

`boolean check = isPrime(15);` `System.out.println(check);`	`false`

4. Write a method called **isPerfect** which takes an integer in its parameter to check whether a number is perfect or not. If the number is perfect then the method returns boolean **true** otherwise it returns boolean **false**.

Sample Input	Sample Output
`boolean check = isPerfect(6);` `System.out.println(check);`	`true`
`boolean check = isPerfect(33);` `System.out.println(check);`	`false`

5. Write a method called **calcTax** that takes 2 arguments which are **your age** then **your salary**. The method must calculate and **return** the tax as per the following conditions:

- No tax if you are less than 18 years old.
- No tax if you get paid less than 10,000
- 7% tax if you get paid between 10K and 20K
- 14% tax if you get paid more than 20K

Sample Method Call	Output	Explanation
`double t = calcTax(16,20000);` `System.out.println(t);`	`0.0`	Here, the age is less than 18 so 0 tax.
`double t = calcTax(20,18000);` `System.out.println(t);`	`1260.0`	Here, the age is greater than 18 and income is between 10K-20K so tax is 7% of 18000 = 1260.

Practice Problems

Strings

Task 1

Trace the following code and write the outputs.

```
public class Trace01{
    public static void main(String[] args) {
        String course = "";
        int i = 2, j = 0, k = 18;
        course = "-->cse";
        while (i< 5) {
            k--;
            j = k;
                while (j > 12 ) {
                if (j % 2 != 0) {
                        course += "<--";
                        course =  course + i + (j / 2);
                } else {
                    course += "-->";
                        course =  course + (i % 2) + j;
                }
                System.out.println(course);
                  if (j == 14) {
                    course = "-->cse";
                }
                --j;
            }
            i++;
        }
    }
}
```

Task 2

Write a method **modifyStrings()** that takes in three given strings **S, S1, and S2** consisting of different numbers of characters respectively, the task is to modify the string **S** by **replacing** all the **substrings S1** with the **string S2** in the string **S** and printing the modified string S.

Sample Input	Sample Output	Explanation
S = "abababa" S1 = "aba" S2 = "a" modifyStrings(S, S1, S2);	aba	Changing the substrings **S[0, 2]**(Referring to characters from the 0th index of **S** till the 2nd index of S and **S[4, 6]** (= S1) to the string S2 (= "a") modifies the string S to "aba". Therefore, print "aba".
S = "baddadda" S1 = "dd" S2 = "n" modifyStrings(S, S1, S2);	banana	Changing the substrings **S[2,3]**(Referring to characters from the 2nd and 3rd index of **S**) and **S[5, 6] (= S1)** to the string **S2 (= "n")** modifies the string S to "banana". Therefore, print "banana".

Strings + Arrays

Task 3

Given an array of email addresses, print the number of valid email addresses satisfying the following conditions.

a) Each email contains an '@' character
b) There is at least one character before and after '@' character and it has to start with letter
c) There is a '.' character after the character(s) after '@' character
d) There is at least one character after '.' character

Sample Input	Sample Output
email_list = {"abc@gmail.com", "!@cv.bd", "123cse@bracu.ac.bd"}	1
email_list = {"cse110@gmail.com", "government@cv.", "eee@bracu.ac.bd"}	2

Task 4

Write a method called isHappyNumber which takes an integer in its parameter to check whether a number is a happy number or not. If the number is a happy number then the method returns boolean true otherwise it returns boolean false. In number theory, a happy number is a number which eventually reaches 1 when replaced by the sum of the square of each digit. For instance, 13 is a happy number because $1^2 + 3^2 = 10$ and $1^2 + 0^2 = 1$. On the other hand, 4 is not a happy number because the process continues in an infinite cycle without ever reaching 1. Unhappy number ends in a cycle of repeating numbers which contains 4 .

Sample Input	Sample Output
boolean check = isHappyNumber(82) System.out.println(check)	true
boolean check = isHappyNumber(4) System.out.println(check)	false

Task 5

Write a method called toDecimal which takes a binary number as a string in its parameter to convert the binary number to its decimal number and return the decimal value. After returning the decimal value, write another method called toHex which takes the converted decimal value in its parameter and calculates the hexadecimal value and then return the hex value.

Sample Input	Sample Output
int decimal = toDecimal("1010") String hex = toHex(decimal) System.out.println(hex)	"A"

Task 6

Trace the following code and write the outputs.

```
class Trace02 {
    public static void main(String args[]) {
        int[] arr1 = {3, 1, 4, 1, 5, 9, 2};
        int[] arr2 = {10, 20, 30, 40, 50, 60, 70};
        int x = 0, y = 0;
        while (x < arr1.length - 1) {
            arr2[x] = arr1[y] * (x + 1) - arr2[y];
            y = 1;
            while (y <= x) {
                arr2[x] = arr2[x] + arr1[y] - y;
                y = y + 1;
            }
            System.out.println(arr2[x]);
            x = x + 1;
        }
        System.out.println(arr2[arr1.length - 1]);
    }
}
```

Task 7

You are given an integer array. You need to identify all the **prime numbers** and **perfect numbers** within the array and print the **indices** along with these **numbers** from the original array.

Sample Input	Sample Output
Sample Input: int arr[] = {6, 13, 28, 17, 3, 9, 11, 23, 10, 29, 12, 7}	Prime Numbers: 1: 13 3: 17 4: 3 6: 11

	7: 23 9: 29 11: 7 Perfect Numbers: 0: 6 2: 28

Task 8

Trace the following code and write the outputs.

```
public class tracing1 {
        public static void main(String[] args){
        int i = 1;
        int [] a = {5,6,7,8,9};
        while (i <= 5){
        int j = a[i%a.length];
        while (j > 1){
                System.out.print(j--);
                if (j == 2)
                break;
        }
         System.out.println("***");
        ++i;
        }
        double x = 7;
        double y = 8;
        double z = 9;
        System.out.println(x < y || y > z);
        System.out.println(x < y && y > z);
        System.out.println(x < y);
        System.out.println(x + y < z);
        System.out.println((x + y)-6 < z);
        }
}
```

Task 9

Your professor expects only As, Bs, and Cs. In the following program, write a method called **getScores** that takes as input corresponding arrays **studentGrades** and **studentScores**. Write a method called **getScores** that assigns **index i** in studentGrades based on **index i** in **studentScores**. If a grade is **A**, assign **100**. If a grade is **B**, assign **90**. If a grade is **C**, assign **70**. If a grade is anything else, assign **0**.

Sample Input	Sample Output
char[] studentGrades = new char[]{'A', 'A', 'A', 'B', 'C', 'U', 'Z'}; int[] studentScores = new int[7];	Output expectation: 100 100 100 90 70 0 0

Task 10

A. Write a method called **convertToCm()**, that takes as input a **type double** and **returns** the value converted from inches to centimeters.
Hint: There are 2.54 centimeters in an inch

Sample Method Call	Output
double t = convertToCm(16); System.out.println(t + " cm");	40.64 cm

B. Create an **array** of **type double** of length **5** called **cheetos_inches**, that stores the length of each of the Cheetos **from the user**. Send the array of length in inches into a method called **findAvgCm()** that **returns** the average length of the Cheetos **in cm to 2 decimal places**. The method findAvgCm() uses **convertTocm()** to convert the length of each Cheetos **from inches to cm.**
Note: You must call the method written in [Method Task A], otherwise this task would be considered invalid.

Sample Method Call	Output
Sample array: double [] cheetos_inches = new double[]{10.0, 12.0, 14.0, 16.0, 18.0}; averageLength = findAvgCm(cheetos_inches); System.out.println("The average Cheeto length is "+ averageLength +" cm");	The average Cheeto length is 35.56 cm

Task 11

A. Write a method called **isVowel** which takes a string in its parameter and counts all the vowels in the String. If any vowel exists in the string then the method returns the **count**.

Sample Input	Sample Output
The quick brown fox jumps over the lazy dog	Number of vowels in the string: 11

B. Write a method called **isConsonant** which takes a string in its parameter and counts all the consonants in the String. If any consonant exists in the string then the method returns the **count**.

Sample Input	Sample Output
The quick brown fox jumps over the lazy dog	Number of consonants in the string: 24

C. Write a method called **vowel/consonantSum** which takes an array of strings in its parameter and returns the summation of the number of vowels/consonants.
Note: **You must call** the methods written in tasks **A/B**, otherwise this task will be **considered invalid**.

Given Array	Sample Output
String [] names = {"Bob", "Alice", "Max", "Marry", "Rosy"}; System.out.println("The total number of vowels in the array is:" + vowelSum(names)); System.out.println("The total number of consonants in the array is:" + consonantSum(names));	The total number of vowels in the array is: **7** The total number of consonants in the array is: **13**